

Notation: $\mathbf{1}$ is the all-ones vector, $\mathbf{e}_i$ is the i -th unit vector, $\mathbf{x}_{p:q}$ is the subvector $(x_p, \dots, x_q)$, and $\|\cdot\|$ is the Euclidean norm.

1. [CO1]

Variant The second-difference matrix

Let $\mathbf{x} = (x_1, x_2, x_3, x_4, x_5, x_6)$ and

$$\mathbf{L} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ -2 & 1 & 0 & 0 \\ 1 & -2 & 1 & 0 \\ 0 & 1 & -2 & 1 \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

- State the size of $\mathbf{x}^\top \mathbf{L}$ and compute it. 3 Marks.
- Write the algebraic equation for the k -th entry of $\mathbf{x}^\top \mathbf{L}$. 2 Marks.
- Express $\mathbf{x}^\top \mathbf{L}$ using only subvectors (slices) of $\mathbf{x}$, with no matrix at all. 2 Marks.
- Evaluate $\mathbf{x}^\top \mathbf{L}$ for $\mathbf{x} = (1, 4, 9, 16, 25, 36)$. Then find a nonzero $\mathbf{x}$ for which $\mathbf{x}^\top \mathbf{L} = \mathbf{0}$, and say in words what $\mathbf{L}$ “detects” in a sequence. 3 Marks.

Variant The moving-average matrix

Let $\mathbf{x} = (x_1, \dots, x_7)$ and

$$\mathbf{A} = \frac{1}{3} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}.$$

- State the size of $\mathbf{x}^\top \mathbf{A}$ and compute it. 3 Marks.
- Write the algebraic equation for the k -th entry of $\mathbf{x}^\top \mathbf{A}$. 2 Marks.
- Express $\mathbf{x}^\top \mathbf{A}$ as a combination of three subvectors of $\mathbf{x}$. 2 Marks.
- Evaluate $\mathbf{x}^\top \mathbf{A}$ for $\mathbf{x} = (2, 2, 2, 8, 2, 2, 2)$ and describe what happened to the spike. Then find a nonzero $\mathbf{x}$ with $\mathbf{x}^\top \mathbf{A} = \mathbf{0}$. 3 Marks.

2. [CO2]

Variant Storage, linear combinations and the Gram matrix

Let $\mathbf{x}_i \in \mathbb{R}^{250}$ for $i = 1, \dots, 400$. Each number is stored in double precision (8 bytes).

- Deduce how many bytes it takes to store the 400 vectors. Give the answer in MB. 2 Marks.
- Count the flops needed to form $\beta_1 \mathbf{x}_1 + \dots + \beta_{400} \mathbf{x}_{400}$ with all $\beta_i \neq 0$. Show the count of multiplications and additions separately. 3 Marks.
- Count the flops needed to compute *every* inner product $\mathbf{x}_i^\top \mathbf{x}_j$ with $i \leq j$. Explain why you need not compute the pairs with $i > j$. 3 Marks.

- (d) On a machine that runs at 5 Gflop/s, estimate the time for (b) and (c). Which task dominates, and by roughly what factor? 2 Marks.

Variant Dense versus sparse documents

A corpus has 5000 documents. Each is represented by a word-count vector in $\mathbb{R}^{20000}$ (one entry per dictionary word). A number takes 4 bytes.

- (a) Deduce the storage needed if every entry of every vector is stored (dense storage). 2 Marks.
- (b) No document contains more than 200 distinct words. If only the nonzero entries are stored, each as a (value, index) pair of 4 bytes each, what is the storage now? 3 Marks.
- (c) Count the flops needed to compute the average document vector $\frac{1}{5000}(\mathbf{x}_1 + \dots + \mathbf{x}_{5000})$ with dense vectors. 3 Marks.
- (d) Estimate the time for (c) at 2 Gflop/s. Estimate it again when the sum uses only the nonzero entries. What principle does the comparison show? 2 Marks.

3. [CO2]

Differentiate between each pair of concepts or operations. Also state what each one means in ordinary English, and support your answer with the example asked for.

Variant A

- (a) $\text{rms}(\mathbf{x})$ and $\text{std}(\mathbf{x})$. Show that $\text{rms}(\mathbf{x})^2 = \text{avg}(\mathbf{x})^2 + \text{std}(\mathbf{x})^2$, and use it to state when the two are equal. 4 Marks.
- (b) The inner product $\mathbf{x}^\top \mathbf{y}$ and the angle $\angle(\mathbf{x}, \mathbf{y})$. Why can $\mathbf{x}^\top \mathbf{y}$ be large while the angle is also large? 3 Marks.
- (c) $\|\mathbf{x} - \mathbf{y}\|$ and $\|\mathbf{x}\| - \|\mathbf{y}\|$. Give a pair $\mathbf{x} \neq \mathbf{y}$ for which the second is 0 but the first is not, and state the inequality that always links them. 3 Marks.

Variant B

- (a) The de-meaned and the standardized versions of $\mathbf{x}$. Compute both for $\mathbf{x} = (1, 3, 5, 7)$ and state the average and standard deviation of each result. 4 Marks.
- (b) The inner product $\mathbf{x}^\top \mathbf{y}$ and the correlation coefficient $\rho(\mathbf{x}, \mathbf{y})$. What happens to each when $\mathbf{x}$ is replaced by $3\mathbf{x} + 5 \cdot \mathbf{1}$? 3 Marks.
- (c) $\mathbf{e}_i^\top \mathbf{x}$, $\mathbf{1}^\top \mathbf{x}$ and $\frac{1}{n} \mathbf{1}^\top \mathbf{x}$, for $\mathbf{x} \in \mathbb{R}^n$. Write the operation that returns the sum of the last k entries of $\mathbf{x}$ as a single inner product. 3 Marks.

4. [CO1]

Write each quantity or condition in vector notation, using only $\mathbf{1}$, inner products, norms and subvectors. Then explain in one line why your expression works.

Variant Shopping baskets

A store sells 30 products. The purchase vector $\mathbf{p} \in \{0, 1\}^{30}$ of a customer has $p_i = 1$ if the customer bought product i and $p_i = 0$ otherwise. A second customer has purchase vector $\mathbf{q}$, and $\mathbf{c} \in \mathbb{R}^{30}$ holds the product prices.

- (a) The number of products bought by *both* customers. 2 Marks.
- (b) The number of products bought by *exactly one* of the two customers. Give two different expressions and show they agree. 3 Marks.
- (c) The amount customer $\mathbf{p}$ spent on products 21 to 30. 2 Marks.
- (d) The condition “customer $\mathbf{p}$ bought every product that customer $\mathbf{q}$ bought”. 3 Marks.

Variant Comparing two patients

Two patients have symptom vectors $\mathbf{s}, \mathbf{t} \in \{0, 1\}^{20}$, where $s_i = 1$ means that symptom i is present. Symptoms 1–8 are respiratory; symptoms 9–20 are not. A severity vector $\mathbf{w} \in \mathbb{R}^{20}$ gives a weight to each symptom.

- (a) The number of symptoms the two patients share. 2 Marks.
- (b) The fraction of symptoms that the patients share, out of all the symptoms that at least one of them has (the *Jaccard similarity*). 3 Marks.
- (c) The condition “patient $\mathbf{s}$ has more respiratory than non-respiratory symptoms”. 2 Marks.
- (d) The total severity of patient $\mathbf{s}$ ’s respiratory symptoms, and the condition “every respiratory symptom of $\mathbf{s}$ is also present in $\mathbf{t}$ ”. 3 Marks.